Eric Cheney

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Education

California Polytechnic State University, San Luis Obispo – BS in Mechanical Engineering May 2027
Formula SAE Battery Systems| ASME | ASHRAE | Transfer Jump Start Program |*GPA:* 3.50/4.00

Evergreen Valley College – AS in Mechanical Engineering May 2025

Experience

Thermal Engineering Intern, Cisco – San Jose, CA June 2026 – August 2026

- Executed thermal interface material (TIM) tests across products spanning graphene, graphite, phase-change, and metallic materials; analyzed pressure, bond line thickness, thermal resistance, contact resistance, and effective conductivity to rank candidates.
- Evaluated TIM, heatsink, and liquid-cooled cold-plate configurations on a 3,000 W thermal test vehicle to compare heat spreading and high-power package cooling performance.
- Performed flatness testing from 30 C to 120 C on new and legacy high-power test boards, measuring an approximately 40% reduction in thermally induced warpage for the revised design.
- Calibrated thermal test equipment and standardized setup and data-collection workflows to support repeatable material and cold-plate comparisons.
- Designed and prototyped an AI-enabled failure-analysis database, dashboard, and agent for Cisco's Innovation Challenge to consolidate records, surface trends, and support faster root-cause investigation.

Mechanical Engineering Intern, Ames National Laboratory (U.S. DOE) – Ames, IA May 2024 – August 2024

- Developed tooling infrastructure and modular toolbox systems to improve lab workflow and equipment organization.
- Audited metal inventories and thermal system components, identifying inefficiencies and saving over $10K in lab expenses.
- Supported advanced materials research through XRF analysis, thermocouple fabrication, high-temperature furnace operations, and lab safety workflow updates.

Projects

ITherm Cold Plate Design Competition (ASME K-16 / IEEE ITherm) Oct 2026 – Present

- Developing an additively manufactured copper cold plate for a 1.5 kW GPU/HBM thermal test vehicle, scored on peak GPU/HBM temperature reduction under a 4 psi pressure-drop limit.
- Built an Ansys Fluent conjugate heat transfer model, including a custom UDF that maps a 630-cell heat-flux map to within 0.5% of the total load, and used mirror symmetry to halve the mesh size.

Cal Poly Racing - Formula Team, Battery Systems September 2025 – October 2026

- Designed forced-air airbox and plenum for accumulator cooling, balancing flow uniformity, packaging, serviceability, and safety constraints.
- Built Siemens NX CAD models and evaluated multiple airbox/plenum iterations using simplified CFD to support design decisions.
- Coordinated with Electrical and Chassis subteams; delivered a compliant design that passed pre-competition technical inspection.

Skills

Thermal & Analysis: Ansys Fluent, Icepak, TIM testing, TTVs, heatsinks, liquid-cooled cold plates, Simcenter Flotherm
CAD & Engineering: Siemens NX, SolidWorks, Fusion 360, Autodesk Inventor, LTspice, GD&T, metrology
Programming & Data: MATLAB, C++, Excel, AI tools, AI agent prototyping, dashboard/database development
Manufacturing: 3D printing, manual mill/lathe, TIG/MIG welding, fixturing